



Opioid Analgesia - Paediatric Procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Health Service	Post Anaesthesia Unit (PACU) / Recovery Room	Medical and Nursing

2. Purpose

For post-operative patients within the Post Anaesthesia Care Unit (PACU) having moderate to severe pain, in order to receive efficient and effective postoperative pain management and to reduce their pain to a comfortable level.

This is achieved by administering intravenous opioid in addition to other multimodal analgesia, administered in incremental doses titrated to the patient's pain score, level of sedation and vital signs.

Applicable to:

This procedure is only for post-operative paediatric patients in the PACU / Recovery Room within the age limit of 6 months to 16 years and having body weight less than 50kg.

If the patient age is <6 months or age is between 16–18 years or weight >50kg, the Anaesthetist in charge will direct either to follow this guideline or formulate an individualised plan or to follow the Opioid Analgesia – Adult Procedure.

3. Equipment

- Opioid as per order on the paediatric Short Stay Medication Chart (AKMR172) in line with the relevant pain protocol
- Sodium Chloride 0.9%, syringes (1, 2, 5, 10 and 20ml)
- Drawing up needle (blunt 18 G needle)
- Appropriate labelling material – refer to the [National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines](#)
- Alcohol swabs
- Kidney dish
- Pictures, cards, and laminated flow charts for pain assessment (e.g., FLACC, Faces), drug preparation and administration guides.

4. Prescription

- The prescription is to be written on the back of the Paediatric Short Stay Medication Chart (AKMR172) in the “As required PRN Medicines” section.
- The order must at a minimum include the following:

Before referencing this procedural document please ensure that you have the latest version from the [Policy Library](#) information hub page.

- Name of drug
- Route
- Dose as per this protocol (e.g., “Fentanyl 2.5microgram bolus as per Recovery Room Paediatric Pain Protocol” – for a patient weighing 10kg).
- Unusual patient conditions and circumstances may lead to practice diverging from this procedure. This should be guided individually by the responsible Consultant Anaesthetist, Duty Anaesthetist or a health care provider delegated by them.

Contraindications

Do not start the pain protocol if the patient:

- Is over sedated with a sedation score of 2 or above
- Has respiratory depression

Age	Respiratory depression
6–12 months old	Less than 20 breaths per minute
1–12 year old	Less than 15 breaths per minute
12–16 year old	Less than 12 breaths per minute

- Has a history of allergic reaction to the prescribed drug or severe anaphylactic reaction to the opioid
- Is hypotensive, hypovolaemic, bradycardic or has poor respiratory effort.

Assessment of the patient and review of the protocol by the attending Anaesthetist, Duty Anaesthetist or an Anaesthetist delegated by them is mandatory in these situations prior to commencement of the pain protocol.

5. Patient assessment tools in recovery before commencing the opioid protocol

The following paediatric pain assessment tools and their suitability of use in different age groups is provided as a guide only. Always confirm with the Anaesthetist in charge if needed.

5.1 Faces, Legs, Activity, Cry and Consolability (FLACC) scale

This tool is aimed at children from 2 months to 7 years. The FLACC scale indicates five identifiers for reporting a child's behaviour that are indicative of pain. A child is assessed under these identifiers and is awarded a score between 0–2 for each area and added to receive a final score (where no pain = 0 and worst pain = 10).

This tool does not require self-report from the child or parent. The FLACC scale is also used in child who can't verbalise or unable to communicate.

FLACC SCALE				
Categories	0	1	2	Total
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, disorientated	Frequent to constant frown, quivering chin, clenched jaw	
Legs	Normal position or relaxed	Uneasy, restless, tense	Kicking or legs drawn up	
Activity	Lying quietly, normal position, moves easily	Squirming, shifting back and forth, tense	Arched, rigid or jerking	
Cry	No Cry (awake or asleep)	Moans or whimpers, occasional complaints	Crying steadily, scream or sobs, frequent complaints	
Consolability	Content, relaxed	Reassured by occasional touching, or being talked to; distractible	Difficult to console or comfort	

5.2 Wong-Baker faces scale

Wong -Baker faces scale can be used for children between the ages of 3–7 years.

Explain to the child that each face represents a person who has no pain (hurt), or some, or a lot of pain. Ask the child to choose the face that best represents the pain they are experiencing. The reliability and validity of this tool in practice is based on telling the child that they do not need to be smiling to have no pain and do not need to be crying to have the worst pain.



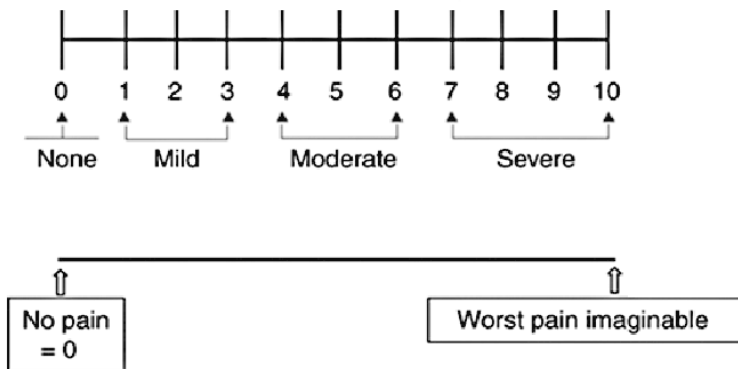
©1983 Wong-Baker FACES Foundation. www.WongBakerFACES.org

5.3 Visual Analogue Scale (VAS)

The VAS is suitable for children over 7 years.

It consists of a horizontal line 10cm long which represents the extremes of pain, where 0 = no pain and 10 = worst pain. When assessing a child's pain, the child is asked to point with their finger to the number between 0 and 10 which indicates their level of pain.

Visual Analogue Scale (VAS)



5.4 Numerical Rating Scale (NRS)

The NRS is suitable for older children above the age of 10 years.

- In the NRS, the child is asked to rate their pain from 0–10 with the understanding that 0 is equal to no pain and 10 is equal to the worst possible pain.
- A score of 1–3 can be considered mild pain, 4–6 as moderate pain and 7–10 as severe pain.

5.5 Sedation score

Sedation Score	
0	Awake and alert
1	Mild - occasionally drowsy, easy to rouse
2	Moderate - constantly drowsy, easy to rouse
3	Severe – somnolent, difficult to rouse
S	Normal sleep, easy to rouse

5.6 Normal Physiological Range for Paediatrics (PALS normal values)

The following tables provide a guide to the normal physiological ranges for paediatric patients:

Age Range	Normal Physiological Range
Respiratory Rate	
0–12 months	30–60 per minute
1–3 years	24–40 per minute
4–5 years	22–34 per minute
6–12 years	18–30 per minute
13–18 years	12–16 per minute

Age Range	Normal Physiological Range	
Heart Rate		
0–3 months	80–200 per minute	
4 months–2 years	75–190 per minute	
2–10 years	60–140 per minute	
Over 10 years	50–100 per minute	
Blood Pressure	Systolic	Diastolic
4–6 months	80–105mmHg	45–70mmHg
7–12 months	70–105mmHg	20–60mmHg
2–6 years	70–105mmHg	25–65mmHg
7–14 years	79–115mmHg	38–78mmHg
15–18 years	90–131mmHg	45–85mmHg

5.7 Checklists before starting pain protocol

On arrival of the patient into the recovery room, the recovery staff must check medication orders with the Anaesthetist in charge and discuss the following measures:

- IV or oral Paracetamol up to 1g if no paracetamol has been given in previous 4–6 hours and is suitable for administration
- IV Parecoxib up to 40mg if no non-steroidal anti-inflammatory (NSAID) has been given in previous 6–12 hours or if no cyclooxygenase-2 inhibitor has been given in previous 12–24 hours and is suitable for administration
- Appropriate opioid analgesia recovery protocol if the patient is expected to have pain requiring opioids in the recovery room.

If the patient is in severe pain upon arrival to recovery, administration of appropriate analgesia beyond the protocol may be carried out and documented by the Anaesthetist in charge.

Prior to commencing pain protocol, the RN must assess the patient for the following:

- **Pain** – The location, cause, and severity of pain (appropriate to the surgery) using appropriate pain scale 0–10 or descriptors the patient can understand: mild, moderate or severe.

Pain assessment may require the use of one more tool. The opioid pain protocol is started only if the pain score is equal to or greater than 4; or if the patient complains of moderate or severe pain and the following requirements are met.

- **Sedation** – The patient must be fully awake or sedated but easily rousable (sedation score 0–1)
- **Respirations** – Respiratory rate must be:
 - More than 20 breaths per minute if 6–12 months old
 - More than 15 breaths per minute if 1–12-year-old
 - More than 12 breaths per minute if 12–16-year-old
- **Oxygen saturation** (SpO₂) must be >94% with or without supplemental oxygen.

- **Blood pressure and heart rate** must be within 20% the patient's normal limits using the preoperative reading and patient history as a guideline. Inform and discuss with the Anaesthetist if it does not lie within the normal limits before starting the pain protocol.

6. Standard operating procedure

- Check the IV cannula site for patency, swelling, inflammation or extravasation.
- Two nurses authorised to handle S8 drugs are to check the medication order in accordance with the requirements of the [Medication Management Procedure](#) – Administration of S8/S4R Medicines section.
- Draw up the prescribed opioid into a 20 ml syringe and make up to 20 ml with Sodium Chloride 0.9% as per [Appendix 1](#).
- Label the syringe with the patient's name, drug and concentration, date and time.
- Refer to the Anaesthetist's order and the Opioid Pain Protocol Flow Chart ([Appendix 2](#)) to calculate the exact dose to administer depending on the patient's age and level of pain score.
- The same two authorised nurses are to check the patient's identification, known allergies and other appropriate patient parameters at the point of administration before commencing the protocol as per the [Medication Management Procedure](#) – Principles of Safe Medication Administration section.
- Provide oxygen therapy as per the Anaesthetist's order.
- Recheck and inject the dose of opioid into the IV cannula port after performing disinfection as per hospital policy as a slow push and flush the line with 0.1ml/kg Sodium Chloride 0.9% or a fluid ordered by the Anaesthetist, drawn up in a separate syringe and labelled. The type and total amount of fluid given is to be recorded on the Intravenous Fluid Treatment and Consent for Transfusion form (AKMR176).
- Ensure appropriate documentation is completed on the Perioperative Nursing Care Plan and Record (AKMR93) and Paediatric Short Stay Medication Chart (AKMR172), including pain score, sedation score, blood pressure, heart rate and oxygen saturation at least 3-minute intervals, indicating the patient's condition prior to each dose and total dose administered. Sedation levels are to be monitored closely and continuously. The pain score is to be assessed prior to each dose of opioid.
- Once the pain protocol is initiated, the patient needs to be monitored continuously and at no time, should the patient or drug be left unattended.
- Re-assess the pain score after every dose and continue with incremental doses until the patient's pain score is 3/10 or less.
- Repeat the dose after a minimum of 3 minutes for fentanyl and pethidine protocol and 5 minutes if morphine protocol.
- If the patient is not comfortable after giving 5 doses of opioid (or maximum dose as per order prescribed), obtain an anaesthetic review of the patient.
- When the patient is ready for discharge from recovery, the two recovery staff members who signed out and prepared the medication are required to witness and document the discarding of any remaining drug which was not used in the appropriate repository in accordance with the [Medication Management Procedure](#) requirements.
- The patient needs to be monitored in the recovery room for at least 20 minutes after the last dose of recovery pain protocol (or at least 30 minutes if greater than 10ml opioid given) despite meeting all other discharge criteria.

- At the time of discharge the respiratory rate must be:
- More than 20 breaths per minute if 6–12 months old
- More than 15 breaths per minute if 1–12 year old
- More than 12 breaths per minute if 12–16 year old
- Sedation score 0–1 and must be easily rousable.
- All staff administering opioid drugs must be aware of their side effects and complications; and must stop the protocol and call the Duty Anaesthetist in charge when one of the following conditions occurs:
- Sedation score ≥ 2
- Respiratory rate < 20 breaths per minute if 6–12 months old; < 15 breaths per minute if 1–12-year-old; or < 12 breaths per minute if 12–16-year-old
- SaO₂ $< 94\%$ with supplemental oxygenation
- Nausea and vomiting unresponsive to anti-emetics
- Blood pressure and heart rate 20% out of normal range
- If the patient is noted to be unresponsive or deteriorates hemodynamically or if the criteria for a [Code Blue – Medical Emergency](#) are met; call for help immediately and activate the Medical Emergency Team (MET) response.

7. Concurrent neuraxial / regional infusion and pain protocol / PCA

- In these circumstances, clearly confirm with the Anaesthetist regarding the drug, doses, and duration in order to avoid medication errors.
- Using the designated colour coded connection tubing and clear labelling is crucial – refer to the [National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines](#)
- Clear and concise handover using the ISOBAR minimum dataset should be completed with the ward staff upon discharge from the recovery.

8. Policy context

The following documents are required to give effect to this procedure:

- WA Health [High Risk Medication Policy MP 0131/20](#)
- WA Health [Medication Review Policy MP 0104/19](#)
- WA Health [Clinical Handover Policy MP 0095](#)
- AKG [Interpreter and Translator Guideline](#)
- AKG [Medication Policy](#)
- AKG [Medication Management Procedure](#)

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.15
- Standard 6 – Action 6.7

National Standards for Mental Health Services (2010):

- Standard 1 – Criteria 1.1

Australia/New Zealand Standards™ (AS/NZS):

- AS/NZS 0000:0000

10. References

1. Hatfield, A., & Tronson, M. (2009). The complete recovery room book (4th ed.). Oxford: Oxford University Press.
2. Simons, J., & Macdonald, L. (2006). Changing practice: Implementing validated paediatric pain assessment tools. *Journal of Child Health Care*, 10(2), 160-176.
3. Baulch, Ingrid. (2010). Assessment and management of pain in the paediatric patient. *Nursing Standard*, 25(10), 35-40.
4. Schug, S. A., Palmer, G. M., Scott, A. D., Halliwell, R., & Trinca, J. Acute pain management: scientific evidence, fourth edition (2016) <http://www.anzca.edu.au/resources/college-publications>
5. Myles, P. S., Myles, D. B., Galagher, W., Boyd, D., Chew, C., MacDonald, N., & Dennis, A. (2017). Measuring acute postoperative pain using the visual analog scale: the minimal clinically important difference and patient acceptable symptom state. *British Journal of Anaesthesia*, 118(3), 424-429. <http://dx.doi.org/10.1093/bja/aew466>
6. International Association for the study of pain (IASP) education material and resources: <http://www.iasp-pain.org/Resources?navItemNumber=650>
7. ACLS & PALS Medical Education & Training; Normal values in Children (as of 18 September 2018). <https://www.aclsmedicaltraining.com/normal-values-in-children/>

11. Document contact

Enquiries relating to this procedure may be directed to:

Title: Consultant Anaesthetist
Department: Anaesthetics

12. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V2	28/09/2023	28/09/2026	Minor amendment

13. Authorisation

This procedure has been authorised and issued by:

Authorised by	Dr Alison Parr - Director Clinical Services
Authorisation date	22/09/2023
Published date	28/09/2023

Appendix 1: Preparing Opioids for PACU/Recovery Room Pain Protocol for Paediatrics

The following instructions for preparation of opioids are to be followed for the adult Recovery Room Pain Protocol Flowchart for Paediatrics ([Appendix 2](#)) in liaison with the above protocol.

Fentanyl protocol

100 micrograms of fentanyl in 2ml diluted with 18ml of sodium chloride 0.9% to make a solution of 20ml containing 5micrograms/ml of fentanyl.

The dose is 0.05ml/kg (or 0.25microgram/kg)

The diluted drug may need to be transferred to either 1 or 2ml syringe.

Morphine protocol

10 mg of morphine in 1ml diluted with 19ml of sodium chloride 0.9% to make a solution of 20ml containing 0.5mg/ml of morphine.

The dose is 0.05ml/kg (or 25microgram/kg)

The diluted drug may need to be transferred to either 1 or 2 ml syringe.

Pethidine protocol

100 mg of pethidine in 2ml diluted with 18ml of sodium chloride 0.9% to make a solution of 20ml containing 5mg/ml of pethidine.

The dose is 0.05ml/kg (or 250microgram/kg)

The diluted drug may need to be transferred to either 1 or 2 ml syringe.

Note:

- Inject the dose of opioid as a slow push and flush the line with 0.1ml/kg of Sodium Chloride 0.9% or specific fluid ordered by Anaesthetist, drawn up in a separate syringe and labelled. The total amount of fluid given is recorded in intake/output and fluid chart.
- The remaining drug in the syringe need to be handled according to the [Medication Management Procedure](#) and kept sterile with sterile caps till the protocol is completed and medication are discarded as per the [Medication Management Procedure](#).
- This preparation guide is based on the current pharmacy supply and may need to re-calculate and modify accordingly if the dose and concentration of supply changes.

Appendix 2: PACU/Recovery Room Pain Protocol Flowchart for Paediatrics

